Bureaucratic Capacity and Urban Planning: Evidence from Los Angeles *

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Abstract

Land use regulation is a central constraint on urban development, yet how regulators actually review development proposals remains a black box. We pry that box open using the complete written record of the Los Angeles City Planning Commission, applying large language models to convert 29,000 pages of agendas, minutes, and public correspondence (2018-2026) into a structured dataset of 818 development decisions. We show that outright denial is rare (1.2% of all cases), and that the more salient margin is delay. We then investigate what features of a case predict delay. Surprisingly, we find that once all factors are controlled for, the physical characteristics of the project have little explanatory power. Instead, public opposition and process-based frictions like how unusual a case is, or when in the agenda order it is heard, matter more. To measure unusualness, we develop a measure of atypicality based on the text embeddings of proposal summaries. We find that a one standard deviation increase in atypicality is associated with a 0.206 increase in the log odds of delay. Notably, the association is stronger when there are fewer members who have seen the case before, and furthermore atypicality only predicts delay, not eventual outcome, thus reinforcing the interpretation that the delays stem from process-based frictions rather than latent proposal quality. The results provide rare field evidence that bureaucratic capacity, not statutory rules alone, shapes the pace of urban development.

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1 Introduction

Modern cities struggle to build housing at the scale required to sustain growth. Rising housing costs have become one of the central economic challenges facing cities in the United States and much of the Western world.[1] A large body of work in urban economics documents that stringent land-use regulations substantially constrain housing supply, contributing to high prices, reduced construction, and spatial misallocation of labor and capital (e.g., Glaeser and Ward, 2009; Turner et al., 2014; Hilber and Vermeulen, 2016; Ganong and Shoag, 2017; Albouy and Ehrlich, 2018; Hsieh and Moretti, 2019; Brueckner and Singh, 2020; Gabriel and Painter, 2020; see Gyourko and Molloy, 2015 and Molloy, 2020 for reviews). The implications extend well beyond housing markets, shaping regional inequality, migration patterns, and aggregate productivity.

Despite this growing literature, most empirical work treats regulation as a static constraint—a set of zoning rules, density limits, or statutory requirements that determine what can be built and where. Much less is known about the *regulatory production process* itself—how development proposals are evaluated, modified, delayed, or approved by administrative bodies, and how this process affects urban outcomes over and above the formal text of land-use regulations. Yet in practice, housing supply is mediated through discretionary review and appeals, which require administrative processing. Even when projects ultimately comply with zoning rules or are eventually approved, conditions introduced during the administrative process can slow the pace of construction. Recent evidence suggests that prolonged approval timelines alone can significantly depress housing production (Gabriel and Kung, 2026), highlighting the importance of bureaucratic capacity as a potential bottleneck in urban development.

[1] Klein and Thompson (2025) have recently brought the issue to popular attention in their book *Abundance*.

This paper studies the regulatory production process in the context of a large urban planning agency. We examine decision-making by the Los Angeles City Planning Commission (LA CPC), the administrative body responsible for approving large development projects in the City of Los Angeles. Commissioners act on behalf of the City, charged with advancing development while ensuring compliance with zoning rules and public expectations. They evaluate proposals combining legal, architectural, and community considerations under time constraints. The public nature of the process generates a rich paper trail: every meeting produces agendas, minutes, and written comments that record both the information presented and the decisions made. We exploit this paper trail to construct a novel dataset of CPC hearing decisions. We leverage large language models to read and encode the documents, converting the over 29,000 pages of CPC documents into a dataset of 818 decisions spanning May 2018 to February 2026, and recording the outcome of each case, the votes cast, the physical, geographic, and planning characteristics of each proposal, and the volume and content of public support and opposition.

A first finding is that outright denial is rare. Out of the 818 cases in our data, only 10 were denied outright. Another 4 were withdrawn, and 354 were approved outright and 328 were approved with conditions or modifications. The remaining 122 were delayed to a future hearing. Out of the 328 approved with conditions, 32 were classified as major conditions while 296 were classified as minor. For projects that were delayed but eventually voted on in our dataset, we observed an average of 105 days between the first delay and the eventual resolution. This is a highly salient amount of time, considering that the average approval time for multifamily projects in Los Angeles is about 525 days.[2] Moreover, over 90% of these projects were eventually approved, most with minor or no changes to the proposal. The findings

[2] See Gabriel and Kung (2026).

suggest that delay, rather than denial or significant project alterations, is the most salient outcome hindering development at the CPC. These delays are likely to have meaningfully large impacts on total real estate development.[3]

This is not to say that regulations have no effect on the mix of projects that get built; they can substantially shape what gets submitted in the first place. Once a project lands on the CPC's desk, however, our findings show that delay is the most salient friction. We therefore turn to investigating the factors that drive approval and delay at the CPC. To do this, we develop and estimate an ordered logit model where the three outcomes are outright approval, approval with conditions, and delay or denial. We are particularly interested in teasing out the marginal influence of the project's physical characteristics, planning characteristics, public support and opposition, and bureaucratic or cognitive frictions. We emphasize the latter because the literature on bureaucratic decision-making has highlighted the importance of both principal-agent problems and bounded rationality in limiting the efficiency of the bureaucratic process.[4] Distinguishing between the influence of physical and planning characteristics from the influence of bureaucratic, political, and cognitive frictions is important because they have very different policy implications. If physical and planning characteristics are the primary drivers of delay at the CPC, then policies to speed up approvals would emphasize reforms to the zoning code and perhaps better project pre-planning. On the other hand, if bureaucratic, political, and cognitive frictions are the primary drivers of delay at the CPC, then policies to speed up

[3] Gabriel and Kung (2026), for example, estimate a large effect of approval duration on housing production, e.g. if multifamily approval times had been 25% faster in Los Angeles between 2010 and 2022, then multifamily housing production would have increased by 12.7% in that period.

[4] On principal-agent, reputational, and incentive issues, see Dixit (2002); Prendergast (2003); Besley and Ghatak (2003); Black (2008); Carpenter and Krause (2012); Damonte et al. (2014); Busuioc and Lodge (2016); Maggetti and Papadopoulos (2018). For bounded rationality, limited attention, and such, see Radner (1993); Van Zandt (1999); Sims (2003); Güth (2010); Salant (2011); Lim et al. (2014); Zhang and Levin (2017); de Clippel and Rozen (2021); Richters (2021); Hébert and Woodford (2023).

approvals would emphasize public outreach, procedural streamlining, and expanding bureaucratic capacity.

After estimating the ordered logit model, we found that four factors were statistically significant and robust in predicting approval and delay. Notably, physical dimensions of the project and development type were *not* found to be statistically significant. First, we found that the amount of public opposition significantly reduces the likelihood of approval and increases the likelihood of delay. A doubling in the number of opposition letters predicts a 0.160 increase in the log odds of delay. Second, we found that the ordering of the item in the agenda significantly influences the chances of approval or delay. A project which is listed one item later in the agenda has a 0.136 increased log odds of approval. This is consistent with both decision fatigue and non-random ordering of agenda items (more controversial items tend to be front-loaded, as a courtesy to the public who may wish to speak on the matter), though we do not disentangle between the two. Third, we found that being placed on the consent calendar significantly predicts approval. An item on the consent calendar was found to have a 1.708 increased log odds of approval. Items are placed on the consent calendar when the chair has deemed an item to be routine and non-controversial. Given that the consent calendar is a procedural tool meant to streamline decision-making, its usage and influence suggests that bureaucratic capacity is a salient constraint faced by the Commission.

Lastly, we found that "unusual" cases are less likely to be approved and more likely to be delayed. To measure how unusual a case is, we develop a measure of atypicality based on how far a case is to its nearest cluster of cases in embedding space. Cases that are "far away" from the nearest cluster are more unusual and cases that are "close" to the nearest cluster are more typical. We describe this procedure in more detail in Section 4. We found that a one standard deviation increase in atypicality increases the log odds of delay by 0.206. The effect is stronger when the share of commissioners who have never heard the case before is larger, and moreover atypicality does *not* significantly predict eventual denial or CPC-mandated alterations to the project, which suggests that unfamiliarity is the driver of delay, not latent proposal quality.

The paper contributes to two strands of the literature. First, it extends research in urban economics on land-use regulation and housing supply by opening the "black box" of the regulatory production process. Our results suggest that the pace of development depends not only on statutory constraints but also on the administrative capacity of the regulatory apparatus itself. Second, the paper contributes to the literature on bureaucratic decision-making. Experimental studies have documented satisficing and cognitive constraints in stylized laboratory settings, but systematic evidence from the field remains scarce, in part because cognitive limits are inherently unobservable. We contribute large-scale, field-based evidence consistent with bounded rationality and oversight risk shaping real regulatory outcomes, along with a scalable, text-based approach to measuring the cognitive dimension of casework that can be applied to other institutional settings where a written record of decision-making exists.

The remainder of the paper proceeds as follows. Section 2 describes the Los Angeles City Planning Commission and its role in LA's planning process. Section 3 describes how we collected and extracted the data used for this research. Section 4 provides descriptive statistics regarding the data. Section 5 investigates which factors are most predictive of CPC outcomes using multivariate regression. Section 6 concludes.

2 Institutional Background

2.1 The Los Angeles City Planning Commission

The Los Angeles planning and approvals process for urban development is a multi-layered process that requires the input of multiple agencies. If a development is "by-right", meaning that it conforms to all zoning regulations within its applicable zone, then development can proceed with a permit from the Department of Building and Safety. If, however, the planned development requires an exception to the zoning code, or has a planned purpose not normally allowed by the zoning code, then approval needs to be obtained from the Planning Department and potentially from the City Planning Commission (CPC).

The LA CPC is a nine member administrative body with members appointed by the Mayor of LA and confirmed by the City Council. The CPC is responsible for reviewing citywide zoning changes, handling approvals and conditional use permits for large developments, and handling appeals of decisions made by lower bodies. The LA CPC is the highest planning body within the City's Planning Department, and sits below only the City Council in the City's overall planning hierarchy. CPC appellate decisions are often final and not appealable, and when they are appealable they may only be appealed to the City Council.

The CPC meets approximately every two weeks. The agenda for each meeting is set by the Commission itself and made available to the public at least seven days before each meeting. To streamline hearings, the chair may place items deemed non-controversial on a consent calendar, which is voted on as a single motion. After each meeting, the meeting minutes are published online. The minutes record the order of discussion, the members present, the motions made on each agenda item, and the votes made on each motion. Motions are passed by majority rule, subject to a quorum of five members present. In addition, members of the public are allowed to submit comments on any agenda item either in writing or in person during the meeting. These procedures determine not only what the Commission decides but the conditions under which it decides: publicly, on a fixed schedule, and with its full docket in view.

2.2 The Commission's Decision Problem

These conditions cast commissioners in a decision environment where they act as agents on behalf of the city. They are appointed rather than elected, they serve part-time, and they are charged with advancing development while ensuring compliance with zoning rules and public expectations. The capacity and conduct of individual bureaucrats in positions like this have measurable consequences for government performance (Decarolis et al., 2020; Fenizia, 2022; Best et al., 2023), and overloading bureaucracies relative to their capacity degrades the quality of implementation (Gratton et al., 2021). The cases that reach the CPC are by construction those that the zoning code does not resolve on its own, so the discretion commissioners exercise is genuine. But they exercise it in public, on the record, and subject to two constraints.

The first constraint is oversight risk. Every agenda, motion, and vote is published. Hearings are attended by residents, community organizations, and representatives of applicants, and CPC decisions can be appealed to or vetoed by the City Council. Oversight of this kind operates through what McCubbins and Schwartz (1984) call fire alarms, in which affected parties raise the alert when a decision threatens their interests. Written public comments are the fire alarms of the planning process, and the people who pull them are disproportionately opposed to new development and effective at delaying them (Einstein et al., 2019; Hankinson, 2018). The resulting exposure is asymmetric. A commissioner who approves a contested project makes an attributable decision, whereas postponing the same project makes no such commitment. None of this requires commissioners to be careless or self-interested. Caution of this kind can be an equilibrium response to the incentives bureaucrats face (Prendergast, 2003).

The second constraint is attention. The Commission meets roughly biweekly and typically faces around ten decision items per hearing, which vary in the interpretive effort they demand. Many cases are routine and look like other cases that the Commission has seen before, and they can be evaluated largely on pattern. Others present questions the Commission has no template for, such as a request to host concerts, film screenings, and charitable fundraisers, with alcohol service, on the grounds of a 53-acre cemetery (a real case in our data, to which we return in Section 4). An unfamiliar case must be worked through from scratch, and the time spent doing so is time not available for the rest of the docket. The consequences of scarce attention in screening bureaucracies are well documented. Patent examiners given less time per application scrutinize less and grant more invalid patents (Frakes and Wasserman, 2017), and professional decision makers ranging from asylum judges to loan officers exhibit systematic cognitive errors when processing long sequences of cases (Chen et al., 2016). Decision makers are boundedly rational (Simon, 1955), allocating scarce attention across competing demands (Maćkowiak et al., 2023). What remains scarce is evidence on how these constraints operate inside administrative bodies, where the cognitive demands of a case are not directly observable.

A body operating under these two constraints would be expected to economize on both attention and exposure. Familiar, uncontested cases should move quickly. Indeed, the Commission has a procedural device for this purpose, the consent calendar, which batches items the chair deems non-controversial into a single vote. How the

Commission handles the rest of its docket—which cases move quickly, which acquire conditions, and which are put off—is not something its rules announce; it must be read off the record of its decisions. The two constraints described here supply the lens through which we read that record, and we return to them when interpreting the results in Section 5.

3 Data Development

3.1 Data Acquisition

The Los Angeles Planning Department's website maintains robust public documentation of City Planning Commission meetings. For each meeting, the agenda, minutes, and any supplemental documents relevant to the meeting (letters from the public, traffic assessments, architectural reports, etc.) are available for download as PDF files.[5] We downloaded these documents for all City Planning Commission meetings from May 10th, 2018 to February 2nd, 2026. This resulted in documentary data for 175 meetings, covering 1,730 agenda items, with 5,874 supplemental documents, spanning 29,131 pages of PDF documents. Download occurred twice: once on April 10th, 2025, and a second time on May 15th, 2026.

Since we are primarily interested in the regulatory approvals process, we limit our attention to the agenda items that require a decision from the board. These are identified by agenda items titled according to their Planning Department case numbers, which have a standardized format of "[CASE PREFIX]-[YEAR]-[SERIAL NUMBER]-[CASE SUFFIXES]". The prefix identifies the decision-making body initially responsible for approval, and the case suffixes identify requested entitlements

[5] As of June 18th, 2026, these documents are available at the URL: `https://planning.lacity.gov/about/commissions-boards-hearings`.

and other case features. Altogether, there were 818 agenda items requiring a decision from the board, as identified by their Planning Department case numbers. The other agenda items include items like "Director's Report" and "General Public Comment" which typically do not require any decision-making on the part of the board.

A typical agenda item is a request from a developer to approve a development plan that goes beyond what the site's zoning designation would allow. Figure 1 shows an example of an agenda item. The case number is DIR-2019-6048-TOC-SPR-WDI-1A. This was a project that was initially approved by the Director of Planning (DIR). The suffixes indicate that the project is utilizing incentives under the City's Transit Oriented Communities (TOC) program, the project requires a discretionary Site Plan Review (SPR), and the project was granted a waiver of dedications and improvements (WDI). However, the decision of the Director of Planning was appealed (1A), and so the appeal and the project proposal are now being considered by the City Planning Commission. In addition to the information contained in the case number, the published agenda contains other data such as the Council District that the project is located in and other specific details about the project proposal.

Figure 2 shows the associated minutes for this agenda item. From the minutes, we can see that the appeal was granted in part and denied in part. The CPC upheld the Director of Planning's previous decision, but additional conditions were applied, thus allowing the project to move forward as long as the developer adheres to the new conditions. This outcome—the partial granting of an appeal and the approval of a project with conditions or modifications—is common but not the only kind of outcome. Sometimes, the applicant's requested actions are granted in their entirety without additional conditions or modifications. Rarely, the requested actions are denied entirely. A more common occurrence than denial is that the CPC puts the decision off to a later date.

Figures 3 and 4 show examples of letters submitted by the public in support of and in opposition to the above project. The support letter emphasizes how the project will ease traffic, reduce air pollution, and increase housing availability. The opposition letter emphasizes concerns about displacement and how the proposed units will be unaffordable to current residents of the neighborhood. These letters typify the kinds of concern expressed by community residents in this dataset; however, the number of letters that this project attracted is atypical, as this happened to be one of the more controversial projects. We will discuss the distribution of support and opposition letters across projects later in Section 4.

3.2 Data Extraction

The documentary data provides a wealth of information about CPC cases and their outcomes. However, the information is locked within textual documents that are difficult to extract using traditional methods. For example, traditional NLP methods based on token and pattern matching would have a hard time comparing the agenda to the minutes and determining whether the requested actions were approved, partially approved, approved with conditions, denied, or whether the decision was postponed to a future date.

To extract usable features from the textual content more robustly, we make use of Anthropic's `claude-opus-4-6` generative language model. The model is used to read the text of the agenda, read the text of the minutes, and to extract structured information about the proposed project and about the decisions of the Commission. Our approach follows a growing literature in economics that uses large language models to impute structured data from unstructured text (e.g., Korinek, 2023; Dell, 2025; Ash and Hansen, 2023; Ziems et al., 2024). The methodology and prompts we used to perform the data extraction are described in detail in the appendix, and are robust to using alternative language models.[6] In this section, we will simply focus on the data features that were extracted from the published texts.

Agenda items. For each agenda item, we extracted the following information from the published agenda:

- Agenda item number (used primarily for identification);
- Item title (for cases requiring a decision, this is always a Planning Department case number);
- Short LLM-generated summary of the agenda item's content;
- The Council District(s) to which the item applies;
- The type of case (residential development, mixed-use development, non-residential development, or other);
- If the case is a development proposal, the physical dimensions of the proposal where applicable (number of units, square footage, height, parking spaces);
- List of requested actions.

Minutes. For each agenda item, we extract the following information from the published minutes text:

- Short LLM-generated summary of the deliberations and the motion that was ultimately voted on;
- Implication of the motion for the proposed project: whether the requested actions were approved, approved with minor conditions and/or modifications, approved with major conditions and/or modifications, denied, or whether deliberations were continued to a future date;

[6] The full data extraction pipeline, including all prompts and scraping code, is also publicly available at `https://github.com/ed-kung/lacpc`.

- Result of the vote (whether the motion passed or failed);[7]
- Vote tallies: the number of ayes, nays, absences, abstentions, and recusals.

Supplemental documents. For each supplemental document, we extract the following information from its text:

- Type of document: whether it is a public comment expressing support or opposition to the case, or whether it was technical documentation attached to the project or an administrative or procedural document;
- Which agenda item(s) it references;
- Short LLM-generated summary of the document contents;
- Support or opposition: For each agenda item that the document references, whether the document supports, opposes, or is neutral towards the development proposal in the referenced agenda item(s);
- Reasons for support or opposition: For each referenced agenda item that the document supports or opposes, a list of mentioned reasons for the support or opposition.[8]

4 Data Description

The resulting dataset contains 818 agenda items with motions voted on by the CPC. Because the dataset is novel, we devote an entire section to describing and exploring the data. These descriptive results may be useful for future researchers who want to work with similar data, or for policymakers who want to better understand the

[7]Note that a motion passing is not the same as the project getting approved, nor is a motion failing the same as the project getting denied. A Member may move to deny the project's requested actions, or move to accept an appeal of a previously approved project, in which case the motion passing implies a denial of the project. These nuances highlight why LLMs are helpful in the data extraction process.

[8]The list of possible reasons that the LLM was asked to select from are shown in Table 2.

current planning process in Los Angeles.

4.1 Distribution of project outcomes

Table 1 shows the distribution of the project outcomes by the unanimity of the vote. Two important facts emerge about the CPC process. First, a minuscule number of projects are denied outright (1.2% of all cases). Rather than being denied, a more common occurrence is that the decision is postponed to a later date (14.9% of cases) or the requested actions are approved with major or minor conditions or modifications (3.9% and 36.2% respectively).[9] 43.3% of the cases were approved without any further conditions or modifications.

Second, most decisions were unanimous (91.8% of all cases). Because of these two facts, we do not view disagreement *within* the board as a significant source of friction in moving development projects through the pipeline. Moreover, few projects are denied outright, so the impact of CPC hearings on final outcomes must come through either i) a slowing down of the process due to having to wait for the decision, which itself may be delayed multiple times; or ii) CPC-mandated alterations that potentially add cost or time to the development, or which could dissuade the developer from even moving forward with the project. Our primary analysis will therefore focus on the bureaucratic factors which lead to either delayed decision making or attaching conditions to the project proposal.

[9] Examples of minor conditions and modifications include minor adjustments to the approved Floor Area Ratio, minor changes in number of approved parking spaces, and adding required security camera locations. Examples of major conditions and modifications include major changes to the allowed number of dwelling units, increases to the required number of affordable set-asides, and changes to the requested zoning designation.

4.2 Distribution of public comments

Of the 5,874 supplemental documents included in the dataset, 5,020 were identified as public comments. 2,243 expressed support for one or more project proposals on the agenda to which they were submitted, and 2,712 expressed opposition to one or more project proposals.[10] The average number of public comments received for the 818 cases in our analytical dataset was 7.8 letters per case.[11] Figure 5 shows the distribution of the number of letters received per case, in support and in opposition. The distribution is thick-tailed with the majority of cases receiving under 5 letters, whereas a smaller number of controversial cases receive the lion's share of public comments.

Table 2 documents the reasons offered for support or opposition in public comments. For each letter supporting or opposing an agenda item, the LLM was asked to extract a list of reasons for the support or opposition, chosen from a list of possible categories. Table 2 shows the possible categories, and the number of letters mentioning each of those categories, grouped separately by support letters and opposition letters. Within each group (support or opposition), the categories are listed in order of frequency of mention, thus showing which categories are most salient to the public.

Among support letters, the most frequently mentioned reasons are related to housing supply and affordability, transit access and walkability, and procedural issues. We note that support letters are less likely to be written by individual members of the public, and more likely to be written by representative of developers or by housing advocacy organizations, and thus the reasons for support reflect the priorities of this

[10]The numbers expressing support and opposition do not necessarily add up to the total number of public comments because some letters may be neutral, while others may express support for some agenda items and opposition to others.

[11]The number of letters per case is not simply the number of public comments divided by the number of cases because some letters address multiple projects. We count such letters as separate instances of a public comment for each project the letter references.

group of actors. The category of "procedural issue" relates to questions on whether the developer followed proper procedures in seeking project approval. Support letters mentioning procedural issues are offered in defense of the developer having followed proper procedures.

Among opposition letters, the most frequently mentioned reasons are related to neighborhood character, building dimensions, and procedural issues. Opposition letters are more likely to be written by individual members of the public, and sometimes by neighborhood councils, so the reasons for opposition reflect the priority of this group of actors. When opposition letters mention procedural issues or community engagement, the complaint is usually that the developer did not follow proper procedure or did not engage sufficiently with community concerns.

4.3 Distribution of case characteristics

Table 3 shows summary statistics for selected case characteristics. For continuous variables, the sample mean is shown both for the overall dataset and by case outcome (approved, approved with conditions, or delayed or denied). For boolean variables, the fraction of cases having that feature is shown, both for the overall sample and separately by case outcome. For each variable, a statistical test is also performed to test equality of means by case outcome and the p-value of that test is shown for each variable.[12]

Case Type. 30.8% of the cases were proposals for new residential developments, 32.0% were for new mixed-use developments (meaning some degree of residential), and 15.6% were for new non-residential developments. The remaining 21.5% include cases

[12]For continuous variables, a one-way ANOVA test is performed. For boolean variables, a Pearson's chi-squared test is performed.

such as requests for conditional use permits for existing developments, or proposals to amend the zoning code in various ways, including both Specific Plan and General Plan updates.

Residential cases are more likely to be approved. Although residential cases account for 30.8% of all cases, they account for 37.3% of all approved cases, and only 24.3% of delayed/denied cases. Non-residential cases are less likely to be approved. Although non-residential cases account for 15.6% of all cases, they account for 9.6% of approved cases and 19.1% of delayed/denied cases. There was no statistical evidence that mixed-use or other type cases were systematically more or less likely to be approved.

Physical Characteristics. The average proposed development in our data was 197,309 square feet large and 115 feet tall, and contained 178 dwelling units, of which 27 were deed-restricted affordable units, and 217 parking spaces. Unsurprisingly, larger projects were less likely to be approved outright, and more likely to be approved with conditions or delayed or denied.

Planning Characteristics. We identified the planning characteristics of each project by utilizing the suffixes in the Planning Department case number. Each suffix indicates a feature of the case relevant to the Planning Department. For example, a suffix of SPR indicates that the development requires a Site Plan Review, and a suffix of DB indicates that the development is utilizing Density Bonus incentives. Altogether, there are 77 unique suffixes in the dataset, which we grouped into 9 categories: Site Plan Review, Appeals, Incentive Programs, Conditional Use Permits, Variances/Adjustments/Exceptions, Compliance Review, Code or Plan Amendments, Development Agreements, and Other.[13]

Site Plan Review. This category indicates that the project needed to undergo Site Plan Review. 33.5% of projects had a suffix in this category. Projects requiring Site Plan Review were less likely to be approved outright, and more likely to be approved with conditions.

Appeals. This category indicates that the CPC is hearing an appeal related to a former decision on this project. 31.7% of projects had a suffix in this category. Appealed projects are less likely to have further conditions placed on them; they are more likely to be either postponed or approved as-is.

Incentive Programs. This category indicates that the project is utilizing or requesting to utilize an incentive program offered by the City or State. These include programs like Los Angeles's Transit Oriented Communities program, which allows for streamlined development near public transit, or California's Density Bonus law, which allows for higher density zoning in exchange for affordable unit set-asides. 47.8% of projects had a suffix in this category. Projects utilizing an incentive program were more likely to be approved outright and less likely to be approved with conditions or delayed.

Conditional Use Permits. This category indicates that the development is requesting a conditional use permit. These are permits to use the property in ways that the zoning code does not normally allow for, like operating a school of a certain size, or selling alcohol. 36.9% of projects had a suffix in this category. Projects requesting a conditional use permit were less likely to be approved outright, and more likely to be approved with conditions or postponed.

Variances/Adjustments/Exceptions. This category indicates that the development

[13] Refer to Appendix Tables A1 and A2 for a full list of all 77 suffixes, their meanings, and their assigned categories.

is requesting a zone variance, line adjustment or other exception to the zoning code. 24.9% of projects had a suffix in this category. Projects requesting a variance, adjustment, or exception were less likely to be approved outright and more likely to be approved with conditions or delayed.

Compliance Review. This category indicates that the development had to undergo a review for compliance with rules related to an overlay zone or other special district that applies to the property. 14.1% of projects had a suffix in this category. Projects requiring compliance review were more likely to be approved with conditions, and less likely to be approved outright or delayed, though the differences are not statistically significant.

Code or Plan Amendments. This category indicates that the proposal seeks to make changes to the zoning code or the City's General Plan or an applicable Specific Plan. 19.4% of projects had a suffix in this category. Projects requesting a code or plan amendment were more likely to be approved with conditions, and less likely to be delayed.

Development Agreements. This category indicates that the project involves an agreement between the developer and the City, such as the City granting a waiver of dedication and improvements, or the City allowing one developer to transfer development rights to another developer. 12.8% of projects had a suffix in this category. Projects with development agreements were less likely to be approved outright and more likely to be delayed, though the differences are not statistically significant.

Public Input. The average number of public comments per case was 7.8, of which 3.3 were in support and 3.9 were in opposition. Receiving more letters is associated with greater controversy and a higher likelihood of delay. The average number of letters received by delayed/denied cases was 11.1, whereas the average number of letters received by approved cases was 4.4. The difference is statistically significant at the 1% level. It is interesting to note that delayed cases received both more letters in opposition *and* more letters in support. This suggests that letters of support are often written only in response to the controversy that a case generates.

Hearing Characteristics. We also explored whether logistical features related to the hearing itself mattered for case outcomes. First, we looked at the order of the item in the agenda. The average agenda item requiring a decision from the board appeared as the 7.7th item on the agenda. In these univariate comparisons, there does not appear to be any evidence that the agenda item order matters for case outcome (though this conclusion changes as we move to the multivariate regressions).

Second, we looked at how many items were on the agenda for that hearing. In theory, a busier agenda could mean that cases on that agenda are more likely to get postponed and less likely to be approved with conditions, since adding conditions takes extra time. We do find some evidence of this: cases on fuller agendas are more likely to be postponed and less likely to be approved with conditions, and the difference is statistically significant at the 5% level.

Third, we looked at the influence of an item being on the "consent calendar". The consent calendar is a practical tool for streamlining meetings in which multiple items are voted on as a whole in a single motion. The consent calendar includes items that the Commission chair has deemed non-controversial and therefore not requiring separate discussion. The consent calendar is published before each hearing, and items can be taken off by Commission members before or after the meeting. 15.6% of the cases were voted on as part of a consent calendar, and cases on the consent calendar were significantly more likely to be approved and significantly less likely to be delayed or denied.

Case Atypicality. In addition to logistical features of the hearing, we wanted to test whether the unusualness of a case matters for its outcomes. In theory, highly unusual cases demand more cognitive bandwidth, and possibly more discussion time, from Commission members. Thus, we hypothesize that decisions on highly atypical cases are more likely to be postponed than decisions on run-of-the-mill cases.

To compute a measure of case atypicality, we first calculated the text embeddings of each agenda item using OpenAI's `text-embedding-3-small` embeddings model. A text embedding is a N-dimensional vector representation of a text's meaning.[14] To remove potential differences in formatting, boilerplate, or how individual CPC staff members write agenda items, we first asked `claude-opus-4-6` to provide a short summary of the item, focusing on key development aspects of the case, such as project type, location, and requested actions. This summary was then passed into the embedding model. For each agenda item, the model returned a 1,536 dimensional numerical vector that represents the semantic meaning of the text. Two agenda items with very similar proposals will have embeddings that are close to each other, while two agenda items with very different proposals will have embeddings that are far away from each other.

Because `text-embedding-3-small` was trained on a general corpus of documents, not specialized to the topic of municipal planning and zoning, not all 1,536 dimensions may be relevant for capturing the important differences between our agenda items. We therefore used principal components analysis to extract the 10 linear combinations of these dimensions which explain the most variance in our corpus. Figure 6 shows the scree plot of the principal components analysis. By the 10th principal component, there are diminishing returns to including more components.

[14] For an introduction to the concept of embeddings, see Mikolov et al. (2013) and Le and Mikolov (2014).

After reducing the embeddings down to 10 dimensions, we grouped the agenda items into three clusters using K-means clustering. Identifying clusters in our data is important because the CPC handles many different types of cases, and the language used could be different across different case types. For example, the language used in a case involving the demolition of an existing building followed by the reconstruction of a multifamily home would be different from a request for a conditional use permit to operate a school. Thus, we measure atypicality only within a set of similar case types, as defined by the clustering algorithm.

Figure 7 shows a scatter plot of the agenda items according to their first two principal components, colored by cluster. Cluster 0, the smallest cluster, consists mainly of non-development actions, such as citywide code amendments, plan updates, or conditional use permits on existing properties. Cluster 1 consists primarily of large scale private development projects in which the CPC is the initial approver. Cluster 2 consists primarily of appeals of decisions made by lower level agencies.

To measure the atypicality of an agenda item, we calculated the Mahalanobis distance between the agenda item's 10-dimensional PCA-reduced embedding to its cluster's centroid. We chose Mahalanobis distance because it accounts for the correlation structure of the reduced embedding space (Rencher and Christensen, 2012, p. 93). A simple Euclidean distance assumes that each principal component is uncorrelated and equally informative. By contrast, the Mahalanobis distance rescales and rotates the coordinate system so that correlations between dimensions are removed. This provides a scale-free and cluster-comparable measure of uniqueness. A Mahalanobis distance of 1.0 indicates that the agenda item lies one standard deviation away from its centroid, regardless of which cluster it belongs to or how dispersed that cluster is in the original embedding space.

Using our measure of case atypicality, we can get a sense of which cases are most unusual for each cluster. In cluster 0, the most atypical case was an application for a conditional use permit to operate cultural events such as charitable fundraisers, film screenings, and concerts, and to sell alcoholic beverages on an existing 53-acre cemetery. In cluster 1, the most atypical case was an application to permit development of a 143.5 foot tall, 201,292 square foot multi-disciplinary research facility on the USC Health Sciences Campus. In cluster 2, the most atypical case involved a Writ of Mandamus issued by the Los Angeles Superior Court ordering the City to set aside a previous approval on a 20-unit housing development because it hadn't adequately demonstrated TOC Tier 3 eligibility. However, the Writ did not limit the discretion vested in the City as to how to proceed with the project.

Table 3 shows how the average value of case atypicality differs across case outcomes. The averages themselves are not meaningful, but the differences across outcome are.[15] Table 3 shows a difference of 0.22 between the average atypicality of a delayed/denied case and the average atypicality of an approved case, which means that delayed/denied cases are about 0.22 standard deviations more atypical than approved cases. The difference is statistically significant at the 1% level.

4.4 Summary of findings

To summarize our findings in this section, we found that: i) residential developments were more likely to be approved and less likely to be delayed, whereas non-residential developments experienced the opposite; ii) larger projects were more likely to be delayed or approved with conditions, and less likely to be approved outright; iii) planning characteristics influence case outcomes, but in various non-monotonic ways; iv) cases receiving a higher amount of public scrutiny were more likely to be delayed

[15] The average of the squared-Mahalanobis distance is always equal to the dimensionality of the feature space.

and less likely to be approved; v) cases were more likely to be delayed and less likely to be approved with conditions when the agenda was fuller; and vi) atypical cases were more likely to be delayed and less likely to be approved outright.

The results are consistent with a model of bureaucratic decision making in which time constraints, cognitive constraints, and public monitoring are all relevant factors in determining the outcome. However, the descriptive results presented in this section do not disentangle the partial effects of each feature from the effects of other correlated features. Therefore, we develop and present the results from a multivariate model in the next section.

5 Statistical Model of CPC Decision-Making

5.1 Model

To investigate the partial effects of case characteristics on case outcome, we developed and estimated a statistical model of CPC decision-making. We model each agenda item at a CPC hearing as having three possible outcomes:

0. The project is denied or the decision is postponed;
1. The project is approved with conditions;
2. The project is approved without further conditions.

Each project proposal i is assumed to have a latent quality variable y_i^* which determines the likelihood of the three outcomes. y_i^* is modeled as a linear function of observed project characteristics and bureaucratic factors, $\mathbf{X}_i$, plus an error term ϵ_i:

$$y_i^* = \mathbf{X}_i \beta + \epsilon_i \tag{1}$$

The latent quality of the project proposal determines its outcome at the CPC hearing. Let $y_i \in \{0, 1, 2\}$ denote project i's outcome. The relationship between y_i^* and y_i is as follows:

$$y_i = \begin{cases} 0 \text{ if } y_i^* < \mu_0 \\ 1 \text{ if } \mu_0 \leq y_i^* < \mu_1 \\ 2 \text{ if } \mu_1 \leq y_i^* \end{cases} \tag{2}$$

with $\mu_0 < \mu_1$. ϵ_i is assumed the follow a standard logistic distribution, and therefore the model is an ordered logit model, and the outcomes are monotonic in $\mathbf{X}_i\beta$. The parameters β, μ_0, and μ_1 are estimated by maximum likelihood.

5.2 Estimation Results

To operationalize the model, we include in $\mathbf{X}_i$ the following variables: indicators for whether the project is a residential, mixed-use, or non-residential development, the natural log of the square footage of the project, the height of the project in feet, the log-base-2 of the number of support letters, the log-base-2 of the number of opposition letters, the order in which the item appears on the agenda, the number of agenda items for that date's hearing, whether the item is on the consent calendar, and the case atypicality.[16] In addition, we include an increasingly comprehensive set of dummy variables, including dummies for the case suffix categories, the Council District to which the project belongs, the year in which the case was heard, and the text embeddings cluster the case belongs to.

[16]We chose not to include other physical characteristics such as number of dwelling units, number of affordable units, and parking spaces because they are highly correlated with the other physical dimensions. We also include dummy variables for whether the case is missing height or square footage information.

Table 4 reports the estimated model coefficients for four different specifications. In column 1, the model includes only dummies for case suffix category, but not for Council District, year, or embedding cluster. In column 2, the model adds Council District dummies. In column 3, the model further adds year dummies. In column 4, the model adds embedding cluster dummies. The purpose of the successive inclusion of dummies is to show robustness of the coefficients of interest to the inclusion of further controls. Column 4 is the preferred specification.

Table 4 shows that although physical dimensions were associated with outcomes in the univariate analysis, the association disappears when other factors are included in the regression. The coefficients on height and square footage are neither large nor statistically significant. Similarly, in the univariate analysis there was a statistically significant association between development type and outcomes. But in the multi-variable regression analysis, the association between project type and outcome is no longer statistically significant.

The variables which are revealed to have a statistically significant association with outcomes are: the number of opposition letters, the order of the item in the agenda, whether the case is on the consent calendar, and the case atypicality. Using column 4 as the preferred specification, the results show that: i) a doubling in the number of opposition letters is associated with a 0.160 reduction in log odds of full approval; ii) moving an agenda item one spot down in the agenda is associated with a 0.136 increase in log odds of full approval; iii) being on the consent calendar is associated with a 1.708 increase in log odds of full approval; iv) and a one standard deviation increase in atypicality is associated with a 0.206 decrease in log odds of full approval.

To translate these results into more interpretable numbers, we report the marginal effects of each variable in Table 5. The results indicate that for an "average" case: i) doubling the number of opposition letters is associated with a 3.3% reduction in the probability of full approval, but a 1.4% increase in the probability of approval with conditions, and a 1.9% increase in the probability of delay or denial; ii) moving an item down one spot in the agenda is associated with a 2.8% increase in the probability of full approval, but a 1.2% decrease in the probability of approval with conditions, and a 1.6% decrease in the probability of delay or denial; iii) being on the consent calendar is associated with a 34.9% increase in the probability of full approval, and a 20.7% decrease in the probability of approval with conditions, and a 14.2% decrease in the probability of delay oar denial; iv) a one standard deviation increase in atypicality is associated with a 4.2% decrease in the probability of full approval, but a 1.8% increase in the probability of being approved with conditions, and a 2.4% increase in the probability of being delayed or denied.

Overall, these results suggest that bureaucratic and political factors drive marginal decision-making at the CPC. If the CPC was primarily motivated by approving small residential developments and denying large non-residential ones, then we would expect to see statistically significant coefficients on project type and physical dimensions. Similarly, if the CPC was primarily motivated by strict adherence to the letter of the law in the zoning code, then we would not expect to find statistically significant influence of public sentiment and logistical issues, once planning characteristics are controlled for. Yet, that is what we find.

The signs of the coefficients are consistent with a theory of bureaucratic decision-making under time constraints, cognitive constraints, and oversight risk. Decisions on unfamiliar projects may be postponed because they require greater cognitive bandwidth, which is scarce under time constraints. Contested projects are also more likely to be postponed because they expose Commission members to greater political and reputational scrutiny. By contrast, routine and uncontested projects proceed with little resistance, as reflected by the consent calendar.

5.3 Bureaucratic capacity or latent quality?

We have interpreted the effect of agenda order and atypicality as arising primarily from time constraints and cognitive constraints. That is, we have interpreted their effects as matters of bureaucratic capacity. A natural question that arises is whether their effects could instead be due to lower underlying quality. If atypical proposals are of lower underlying quality, or if items earlier in the agenda are of lower underlying quality, then we can no longer interpret their association with delay as a matter of bureaucratic capacity, but rather as a matter of underyling quality issues with the proposal.

To test this, we ask whether delayed cases are more likely to be eventually denied or have conditions attached to them. If atypical and agenda order were merely proxies for latent quality, we would expect them to predict worse eventual outcomes regardless of whether the case was initially delayed; that is, they should be significantly more likely to be denied or approved with conditions in subsequent hearings. To test this hypothesis, we remove all delay decisions from the estimation sample and estimate a binary logit model in which the outcomes are 1 if the case is approved and 0 if the case is either approved with conditions or denied. The results for this regression are shown in Table 6. In all four specifications, the coefficients on atypicality and agenda order becomes substantially attenuated. The coefficients on atypicality are no longer statistically significant in any of the four specifications, whereas the coefficients on agenda order are only statistically significant in the fourth regression, and only at the 10% level. Thus, atypicality and agenda order have more statistical power in explaining delay than they do in explaining eventual outcome. This suggests that the association of agenda order and atypicality to delay cannot be attributed entirely to poor latent quality.

To further test our interpretation that the results are driven by bureaucratic capacity, we consider whether other measures of unfamiliarity similarly predict delay relative to approval. To do this, we consider cases that are heard more than once within our dataset. For each case heard more than once, we calculate the share of Commission members who were present at the previous hearing who are now at the current hearing. Our hypothesis is that if there are fewer returning members, the case will be less familiar to the present voting members, and thus fewer returning members will be associated with delay. We also hypothesize that this association will be stronger for highly atypical cases. By restricting the estimation sample to only re-heard cases, we severely reduce our sample size to only 124 cases. We are therefore unable to re-estimate the full ordered logit model. Instead, we estimate a binary logit model in which 0 indicates denial or delay and 1 indicates approval or approval with conditions.

Table 7 reports the results. In addition to controlling for the share of returning members, we also include variables for how many days it has been since the case was last heard, and the number of times the case has already appeared in the data. To avoid over-saturating the model, we estimate different specifications with various groups of controls. In all specifications, we include cluster fixed effects, year fixed effects, and Council District effects. In column 2, we include physical characteristics and project type. In column 3, we include suffix group dummies. In column 4, we include the number of support and opposition letters. In column 5, we include agenda order and number of agenda items. We no longer include consent calendar because of a lack of variation. The results show that when there are fewer returning members, the case is more likely to be delayed, and that this effect is stronger for more atypical cases. The interaction term between atypicality and the share of returning members is statistically significant.

5.4 What happens to delayed cases?

We now assess the extent to which delays at the CPC have a meaningful impact on economic outcomes. Gabriel and Kung (2026) showed that approval delays are an important driver of reduced housing production. They estimated that a 25% reduction in approval times would have increased housing production in Los Angeles by 12.7% between the years of 2010 and 2022. Thus, delays at the CPC could have a meaningful impact on real estate development if they add significant time to the overall approvals process.

Table 8 investigates what happens to the delayed cases in our dataset. There were 80 cases which were ever delayed in our dataset.[17] Of these, 73 eventually received a non-delay outcome within our dataset. The average time between first observed delay and final outcome was 105 days. This is an underestimate because a) we don't observe the final outcome for some of the cases (right-censored resolution times) and b) we don't necessarily observe the first hearing (left-censored start times). But even this underestimate of 105 days is economically significant: Gabriel and Kung (2026) report that the median approval time for multifamily housing development projects in Los Angeles from 2010 to 2022 was about 524 days, so 105 days is a meaningful addition. Moreover, it is interesting to note that between the initial delay and final outcome, the majority of the delayed cases (75.3%) did not change in terms of their requests, and some (17.8%) came back with only minor changes. Only 5 of the delayed cases came back with any major changes to their requests; and even among these, only 2 came back with reduced requests, while the other 3 actually expanded their requests. The fact that a delay at the CPC does not usually cause the applicant to substantially modify their requests again suggests that the CPC's decision to delay is

[17]This number differs from the 122 instances of delay reported in Table 1 because in Table 1 counts multiple hearings of the same case as separate units.

not necessarily correlated with the quality of the proposal, or else one would expect the developers to modify the proposal to increase the chance of approval.

5.5 Discussion

Cognitive limits are inherently unobservable. They leave no entry in an agenda or a minute, and can only be inferred indirectly from patterns in observed decisions. Our estimates provide the basis for one such inference. If approval at the CPC hinged primarily on the fundamentals of a proposal, the fundamentals should carry more explanatory weight. Instead, project type and physical dimensions lose their association with outcomes once controls are included, while the variables describing the *process*—public opposition, case atypicality, agenda order, and placement on the consent calendar—retain theirs.

The decision process described in Section 2 offers an interpretation of this pattern, where the opposition penalty reflects oversight risk and the atypicality penalty comes from scarce attention. But an alternative reading is that these variables simply proxy for latent quality. Atypical cases may be atypical because something is wrong with them. Three features of the data resist this interpretation. First, once delayed cases are set aside, atypicality loses its power to predict final outcomes. Atypicality predicts whether a case is delayed, not how it is ultimately decided, supporting a processing friction rather than a quality defect. Second, the association between atypicality and delay is stronger when the case is re-heard by commissioners who had not seen it before. This is more consistent with lack of familiarity driving the result rather than a quality defect. Third, delayed cases overwhelmingly return unchanged: roughly three quarters come back with identical requests and are then resolved. If delay signaled a quality problem, applicants would presumably revise. Instead, for the most part, they simply wait.

Nonetheless, it may be that by the time a case reaches the CPC it has been screened for basic quality, so that the frictions we document operate only at the margin among pre-vetted proposals. We cannot test these possibilities with the data at hand, and we flag them as open questions rather than resolve them. Our claim is modest: decision-making at the CPC is slowed by bureaucratic and political factors that may be orthogonal to the overall planning objectives of the City. Bureaucratic and political delay should accordingly register on the cost side of any cost–benefit analysis of new regulation. This is, of course, known quite generally, but we provide novel empirical evidence and some quantification of the effect.

6 Conclusion

This paper brought structure to the public record of the Los Angeles City Planning Commission, converting eight years of agendas, minutes, and public correspondence into a systematic dataset of 818 development decisions. That record reveals clear patterns. The Commission's operative margin is delay, and delayed cases typically return unchanged after an average of 105 days. Approval odds fall with case atypicality—a text-based measure we developed to capture how unusual a proposal is relative to the docket the Commission typically hears—and with public opposition, whereas public support provides little offsetting influence. Routine items, particularly those bundled onto the consent calendar, proceed with little resistance. These patterns are consistent with commissioners allocating scarce attentional resources across proposals under public scrutiny, prioritizing cases that are familiar or safe. Institutional mechanisms such as the consent calendar function help economize on attention.

The results also have implications for urban development. Land-use regulation remains one of the central bottlenecks in housing supply, yet little is known about the role of the bureaucratic process vis-à-vis the statutory requirements. Our analysis shows that administrative capacity itself may constrain housing production beyond the direct impact of statutory restrictions. The magnitudes are meaningful: the average 105-day delay we observe at the CPC stage is a substantial addition to the 524-day median approval timeline documented for multifamily projects in Los Angeles (Gabriel and Kung, 2026). Negative public pressure clearly shapes outcomes, but cognitive limits—and thus, bureaucratic capacity—also constrain approval of novel or complex projects. The result is a form of institutional lock-in. By favoring projects that are easiest to interpret and defend, the process may reinforce its own expectations, narrowing the range of designs it can evaluate, and thereby limiting innovation in the kinds of development that could meet residents' needs.

More broadly, our framework demonstrates how text can be used to represent the decision architecture within organizations. With such data becoming increasingly available, atypicality provides a scalable measure to study how information is processed, attention is allocated, and discretion is transformed into routines. These mechanisms deepen our understanding of how organizations adapt, and the conditions under which they resist change.

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Figure 1: Example of an Agenda Item

6. **DIR-2019-6048-TOC-SPR-WDI-1A**
CEQA: ENV-2016-273-MND-REC1
Plan Area: Northeast Los Angeles

Council District: 1 – Cedillo
Last Day to Act: 10-08-20
Continued from: 08-13-20
08-27-20

PUBLIC HEARING REQUIRED

PROJECT SITE: 135 – 153 West Avenue 34; 3401 – 3437 North Pasadena Avenue

PROPOSED PROJECT:
Construction, use, and maintenance of a new, five-story, 514,756 square-foot mixed use building with 468 dwelling units, including 66 dwelling units set aside for Very Low Income Households (or 14 percent of the proposed density) and 16,395 square feet of commercial space. The development will be constructed within two phases, and is designed as one building which includes two levels of subterrean parking across the entire site with three structures above that include residential and commercial uses. The structures will be four and five stories tall with a total of 311 automobile parking spaces, 35 short-term and 264 long-term bicycle parking spaces, and a total of 49,152 square feet of open space for residents.

APPEAL:
An appeal of the June 12, 2020, Planning Director's Determination which:

1. Based on the independent judgment of the decision-maker, after consideration of the whole of the administrative record, the Project was assessed in Mitigated Negative Declaration, No. ENV-2016-273-MND adopted on August 22, 2017; and pursuant to CEQA Guidelines 15162 and 15164, as supported by the addendum dated December 2019, no major revisions are required to the Mitigated Declaration; and no subsequent EIR or negative declaration is required for approval of the Project;
2. Approved, pursuant to Section 12.22 A.31 of the Los Angeles Municipal Code (LAMC), a 70 percent increase in density bonus consistent with the provisions of the Transit Oriented Communities Affordable Housing Incentive Program for a Tier 3 project with a total of 468 dwelling units, including 66 dwelling units reserved for Very Low Income (VLI) Households occupancy for a period of 55 years, along with the following two Additional Incentives:
   a. Setback. To permit the use of any or all the yard requirements for the RAS3 Zone in lieu of the [T][Q]CM-2D Zone; and
   b. Transitional Height. To utilize the Transit Oriented Communities transitional height requirements in lieu of those found in LAMC Section 12.21.1 A.10;
3. Approved, pursuant to LAMC Section 12.37 I, a Waiver of Dedication and Improvement for a five-foot dedication and three-foot widening along Avenue 34 and a 15 feet by 15 feet chamfer or a 20 foot radius corner cut along northwest intersection of Avenue 34 and Pasadena Avenue, in order to maintain the existing condition along Avenue 34 and the corner of Avenue 34 and Pasadena Avenue;
4. Conditionally Approved, pursuant to LAMC Section 16.05, a Site Plan Review for the construction, use and maintenance of a new, five-story, mixed-use building with 468 dwelling units, and 16,395 square feet of commercial space in the [T][Q]CM-2D Zone; and
5. Adopted the Conditions of Approval and Findings.

Applicant: [redacted]

Appellant: [redacted]

Staff: [redacted]

Figure 2: Example of a Minutes Item

ITEM NO. 6

DIR-2019-6048-TOC-SPR-WDI-1A
CEQA: ENV-2016-273-MND-REC1
Plan Area: Northeast Los Angeles

Council District: 1 – Cedillo
Last Day to Act: 10-08-20
Continued from: 08-13-20
08-27-20

PUBLIC HEARING HELD

PROJECT SITE: 135 – 153 West Avenue 34; 3401 – 3437 North Pasadena Avenue

IN ATTENDANCE VIA ZOOM WEBINAR/TELECONFERENCE:
[redacted], City Planning Associate, [redacted], City Planner and [redacted], Senior City Planner representing the Department; [redacted] and [redacted], representing the Applicant; and [redacted] representing the Appellant and [redacted], Appellant.

At approximately 10:55 a.m. President Millman announced Commissioner Leung left the zoom webinar/teleconference meeting during Public Comment for this item.

MOTION:
Commissioner Ambroz put forth the actions below in conjunction with the approval of the following Project with modifications, if any, stated on the record:

Construction, use, and maintenance of a new, five-story, 514,756 square-foot mixed use building with 468 dwelling units, including 66 dwelling units set aside for Very Low Income Households (or 14 percent of the proposed density) and 16,395 square feet of commercial space. The development will be constructed within two phases, and is designed as one building which includes two levels of subterrean parking across the entire site with three structures above that include residential and commercial uses. The structures will be four and five stories tall with a total of 311 automobile parking spaces, 35 short-term and 264 long-term bicycle parking spaces, and a total of 49,152 square feet of open space for residents.

1. Find, based on the independent judgment of the decision-maker, after consideration of the whole of the administrative record, the Project was assessed in Mitigated Negative Declaration, No. ENV-2016-273-MND adopted on August 22, 2017; and pursuant to CEQA Guidelines 15162 and 15164, as supported by the addendum dated December 2019, no major revisions are required to the Mitigated Declaration; and no subsequent EIR or negative declaration is required for approval of the Project;
2. Deny the appeal in part and grant the appeal in part of the Planning Director's decision dated June 12, 2020;
3. Approve, pursuant to Section 12.22 A.31 of the Los Angeles Municipal Code (LAMC), a 70 percent increase in density bonus consistent with the provisions of the Transit Oriented Communities Affordable Housing Incentive Program for a Tier 3 project with a total of 468 dwelling units, including 66 dwelling units reserved for Very Low Income (VLI) Households occupancy for a period of 55 years, along with the following two Additional Incentives:
   a. Setback. To permit the use of any or all the yard requirements for the RAS3 Zone in lieu of the [T][Q]CM-2D Zone; and
   b. Transitional Height. To utilize the Transit Oriented Communities transitional height requirements in lieu of those found in LAMC Section 12.21.1 A.10;
4. Approve, pursuant to LAMC Section 12.37 I, a Waiver of Dedication and Improvement for a five-foot dedication and three-foot widening along Avenue 34 and a 15 feet by 15 feet chamfer or a 20 foot radius corner cut along northwest intersection of Avenue 34 and Pasadena Avenue, in order to maintain the existing condition along Avenue 34 and the corner of Avenue 34 and Pasadena Avenue;
5. Conditionally Approve, pursuant to LAMC Section 16.05, a Site Plan Review for the construction, use and maintenance of a new, five-story, mixed-use building with 468 dwelling units, and 16,395 square feet of commercial space in the [T][Q]CM-2D Zone;
6. Adopt the Conditions of Approval, as modified by the Commission, including Staff's Technical Modification dated August 10, 2020 and October 7, 2020; and
7. Adopt the Findings.

Commissioner Perlman seconded the motion and the vote proceeded as follows:

Moved: Ambroz
Second: Perlman
Ayes: Khorsand, Millman, Mitchell
Nay: Mack
Absent: Choe, Leung

Vote: **5 – 1**

MOTION PASSED

Figure 3: Example of a Support Letter

DIR-2019-6048-TOC-SPR-WDI-1A

[redacted] <[redacted]> Mon, Oct 5, 2020 at 7:18 PM
To: cpc@lacity.org, [redacted]

Dear City Planning Commission and/or [redacted]

My name is [redacted] and I am a resident of Lincoln Heights, only a few blocks away from the proposed project on 135-153 West Avenue 34. Due to my job I am unable to attend the City Planning Commission hearing on October 8th in person, but I would like to make a brief comment in support of the project.

Los Angeles direly needs more housing, and this project demonstrates a solution that helps both low income and moderate income residents. I like how this project includes 66 low income units for families but also provides housing for middle income families since the rest of the units are set at market rate. I also like how close it is to the transit station and how there are fewer parking spots than units. I think we need more housing like this in Los Angeles in order to fight the car culture. Additionally, I love how there is specific bicycle parking for the new building. My only recommendation is that fewer parking spots are allotted so that owning a car would be more difficult - as that is the only way we are going to shift this toxic car culture in the city.

I know many of my neighbors are against this project, for reasons I cannot understand. One of the issues they raise is the pollution on site, but they fail to understand that by promoting a car free culture pollution is actually decreased. I also fail to understand their arguments about being residents being pushed out of the neighborhood. I would hold this project to a much higher standard if it were displacing people on the site but that is simply not the case. Lastly, there is concern about parking since this project will bring in more cars. I am a strong follower of the ideals of Donald Shoup, a professor in the Department of Urban Planning at UCLA. He argues, and I agree, that providing "free parking" is actually extremely expensive for society and ought to be done away with. With less parking available, it will force more people to abandon their cars and use the world class Metro system we have here in Los Angeles.

As a resident of the neighborhood I feel this would be a valuable addition to our community, promote a public transit orientated lifestyle, and help alleviate the housing shortage here in Los Angeles. I hope to see your support of this project on October 8th.

[redacted]

Figure 4: Example of an Oppose Letter

135 Avenue 34 Proposed Project

[redacted] <[redacted]> Sun, Oct 4, 2020 at 12:04 AM
To: cpc@lacity.org, [redacted], [redacted]

DIR-2019-6048-TOC-SPR-WDI-1A
ENV-2016-273-MND-REC1

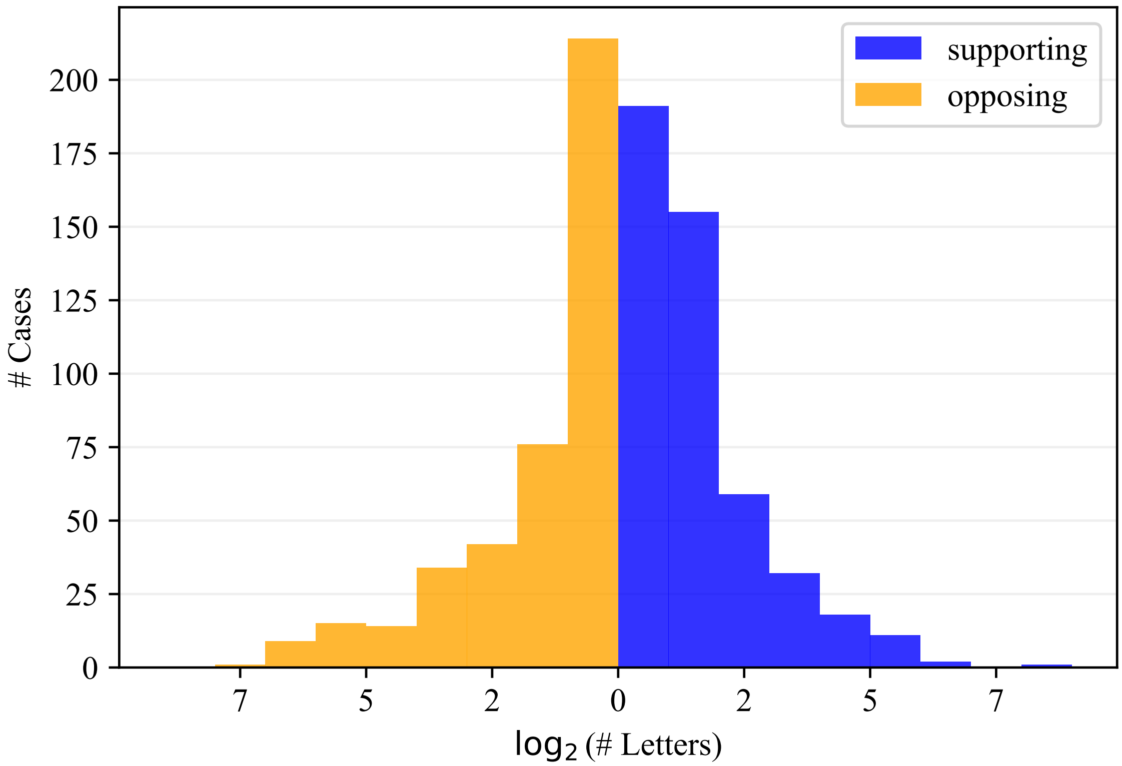
Hello,

My name is [redacted]. I am a resident of Lincoln Heights, and I am emailing today to express my disapproval regarding the proposed development project on Avenue 34 in the Lincoln Heights community.

As an active resident and High School student in the area, this project will affect me because it will drive up neighboring rents, displacing the people who live in the Lincoln Heights community. In addition, this project will create significant displacement. The developers will only include 14% Low-Income units, with 85% at high market rates that will not help the community's current lack of affordable housing. The median combined annual household income in Lincoln Heights is $39,000. This makes even the low-income apartments out of reach for many in our community.
Given my points above, I demand you listen to community feedback and halt this project before it can do any more damage to the community of Lincoln Heights.

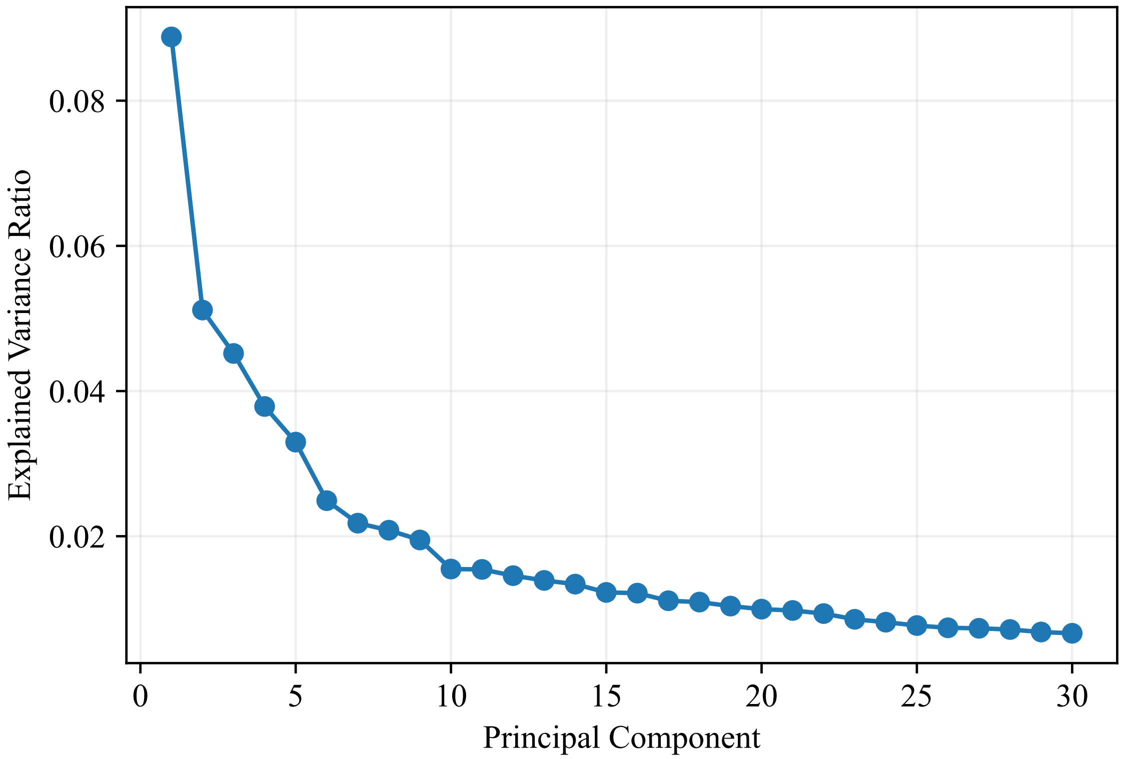
Sincerely,
[redacted]

Figure 5: Distribution of Number of Support and Opposition Letters



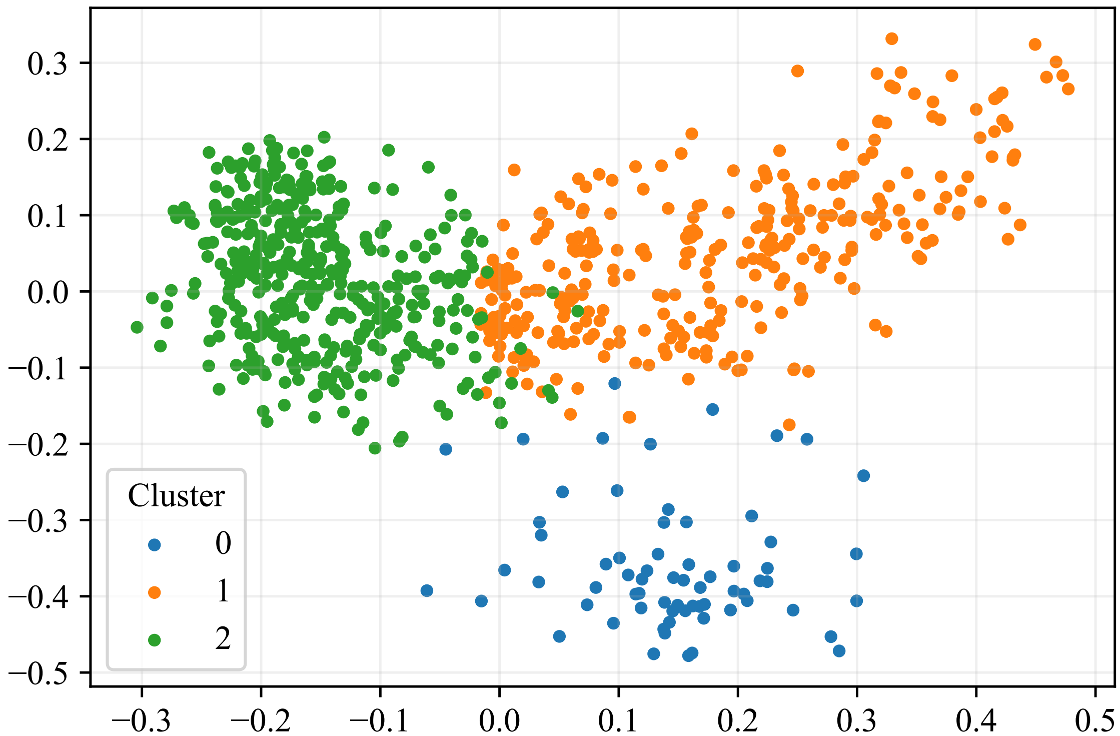
Note: Shows distribution of the number of supporting and opposing letters across cases. Cases with zero support or oppose letters are not included in the figure. Note that one case can have multiple supporting *and* opposing letters.

Figure 6: Scree Plot of PCA Components



Note: Shows how much total variance in the 1,536 dimensional embedding space of the corpus of agenda items is explained by each principal component (up to the first 30).

Figure 7: PCA Reduced Embeddings with K-Means Clustering



Note: Result of K-means clustering with three clusters on the first 10 principal components of the embedding space. The first two dimensions of the 10-dimensional subspace are shown. Each dot is an agenda item.

Table 1: Summary of Motion Outcomes and Vote Results

Project Outcome	Unanimity: Unanimous	1 Nay	>1 Nays	Total	
Approved	333	17	4	354	(43.3%)
Approved with minor conditions	263	21	12	296	(36.2%)
Approved with major conditions	25	2	5	32	(3.9%)
Deliberations continued to future date	119	3	0	122	(14.9%)
Denied	7	1	2	10	(1.2%)
Application withdrawn	4	0	0	4	(0.5%)
TOTAL	751	44	23	818	
	(91.8%)	(5.4%)	(2.8%)		

Notes: This table shows the number of cases decided on by the City Planning Commission, organized by the implication of the motion for the project proposal and the unanimity of the vote.

Table 2: Reasons for Support/Opposition

# Support letters mentioning		# Oppose letters mentioning	
housing supply / affordability	1,387	neighborhood character	1,583
transit access / walkability	824	building height / size / density	1,311
procedural issues	794	procedural issues	1,240
climate / environment	677	community engagement	1,153
displacement / gentrification	631	safety	1,096
neighborhood character	574	traffic / parking	1,065
building height / size / density	568	CEQA related matters	905
economic impacts / jobs	538	housing supply / affordability	866
homelessness	511	climate / environment	832
CEQA related matters	507	displacement / gentrification	577
safety	434	historic / cultural preservation	483
traffic / parking	357	transit access / walkability	278
community engagement	329	economic impacts / jobs	236
historic / cultural preservation	185	homelessness	161
Total support letters	2,243	Total oppose letters	2,712

Notes: This table shows the number of support or oppose letters mentioning each reason. Note that the numbers do not add up to the total number of letters because some letters mention multiple reasons.

Table 3: Descriptive Statistics by Case Outcome

Variable	Overall ($n = 818$)	Approved ($n = 354$)	Approved w conditions ($n = 328$)	Delayed or denied ($n = 136$)	p-value
Case type					
Residential development (%)	0.308	0.373	0.265	0.243	0.002***
Mixed use development (%)	0.320	0.299	0.335	0.338	0.535
Non-residential development (%)	0.156	0.096	0.207	0.191	0.000***
Other case type (%)	0.215	0.232	0.192	0.228	0.420
Physical characteristics					
Square footage (mean)	197,309	131,658	231,355	284,461	0.002***
Height (ft, mean)	115	93	121	155	0.001***
Dwelling units (mean)	178	137	208	229	0.003***
Affordable units (mean)	27	25	34	12	0.106
Parking spaces (mean)	217	108	258	413	0.002***
Planning characteristics					
Site Plan Review (%)	0.335	0.268	0.412	0.324	0.000***
Appeals (%)	0.317	0.336	0.262	0.397	0.010**
Incentive Programs (%)	0.478	0.548	0.436	0.397	0.002***
Conditional Use Permits (%)	0.369	0.325	0.405	0.397	0.071*
Variances/Adjustments/Exceptions (%)	0.249	0.178	0.338	0.221	0.000***
Compliance Review (%)	0.141	0.158	0.137	0.103	0.282
Code or Plan Amendments (%)	0.194	0.175	0.235	0.147	0.045**
Development Agreements (%)	0.128	0.107	0.131	0.176	0.120
Public input					
# Letters (mean)	7.8	4.4	10.0	11.1	0.002***
in support (mean)	3.3	1.9	4.7	3.3	0.027**
in opposition (mean)	3.9	2.1	4.5	7.0	0.001***
Hearing characteristics					
Item agenda order (mean)	7.7	7.7	7.7	7.9	0.459
# Items on agenda (mean)	10.1	10.2	9.9	10.5	0.030**
Consent calendar (%)	0.156	0.234	0.128	0.022	0.000***
Atypicality	3.07	2.99	3.10	3.22	0.007***

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$

Notes: This table reports descriptive statistics for variables grouped by motion outcome. To test differences between groups, a one-way ANOVA F-test was conducted for continuous variables and a chi-square test was conducted for categorical variables. The p-value for such tests is reported in the last column.

Table 4: Factors Influencing Approval and Delay

Ordered Logit Model
0=Delayed/Denied, 1=Approved with Conditions, 2=Approved

	(1)	(2)	(3)	(4)
Residential Development	0.531	0.455	0.558	0.195
	(0.327)	(0.335)	(0.346)	(0.361)
Mixed-Use Development	0.157	0.093	0.152	-0.285
	(0.326)	(0.335)	(0.346)	(0.368)
Non-Residential Development	-0.310	-0.414	-0.326	-0.346
	(0.323)	(0.330)	(0.338)	(0.343)
ln(Square Footage)	-0.035	-0.025	-0.015	-0.053
	(0.059)	(0.061)	(0.064)	(0.064)
Height (ft)	-0.001	-0.001	-0.001	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
$\log_2$(# Support)	0.048	0.049	0.087	0.089
	(0.059)	(0.061)	(0.062)	(0.062)
$\log_2$(# Oppose)	-0.129**	-0.140**	-0.160***	-0.160***
	(0.054)	(0.055)	(0.057)	(0.057)
Agenda Order	0.096**	0.096**	0.111**	0.136***
	(0.047)	(0.048)	(0.049)	(0.050)
No. Agenda Items	-0.026	-0.021	0.006	-0.007
	(0.039)	(0.040)	(0.042)	(0.043)
Consent Calendar	1.550***	1.573***	1.650***	1.708***
	(0.253)	(0.259)	(0.264)	(0.266)
Atypicality	-0.205**	-0.220**	-0.199*	-0.206*
	(0.101)	(0.104)	(0.105)	(0.108)
Suffix Group Dummies	Y	Y	Y	Y
Council District Dummies	N	Y	Y	Y
Year Dummies	N	N	Y	Y
Embedding Cluster Dummies	N	N	N	Y
μ_0	-1.989**	-2.115**	-0.532	-1.883*
	(0.829)	(0.882)	(0.953)	(1.038)
μ_1	0.122	0.038	1.706*	0.387
	(0.825)	(0.879)	(0.956)	(1.035)
Observations	818	818	818	818
McFadden's Pseudo-R^2	0.062	0.074	0.098	0.106

Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Notes: This table reports coefficient estimates from the ordered logit regression described in Section 5. Dummies for missing values of height and square footage are included for cases without reported physical dimensions.

Table 5: Factors Influencing Approval and Delay - Marginal Effects

	(1)	(2)	(3)
	Delay / Denied	Approved with Conditions	Approved
Residential Development	-0.023	-0.018	0.040
	(0.041)	(0.034)	(0.075)
Mixed-Use Development	0.035	0.023	-0.058
	(0.046)	(0.028)	(0.074)
Non-Residential Development	0.044	0.027	-0.071
	(0.046)	(0.023)	(0.069)
ln(Square Footage)	0.006	0.005	-0.011
	(0.008)	(0.006)	(0.013)
Height (ft)	0.000	0.000	-0.000
	(0.000)	(0.000)	(0.000)
$\log_2$(# Support)	-0.011	-0.008	0.018
	(0.007)	(0.005)	(0.013)
$\log_2$(# Oppose)	0.019***	0.014***	-0.033***
	(0.007)	(0.005)	(0.012)
Agenda Order	-0.016***	-0.012***	0.028***
	(0.006)	(0.004)	(0.010)
No. Agenda Items	0.001	0.001	-0.001
	(0.005)	(0.004)	(0.009)
Consent Calendar	-0.142***	-0.207***	0.349***
	(0.017)	(0.033)	(0.046)
Atypicality	0.024*	0.018*	-0.042*
	(0.013)	(0.009)	(0.022)

Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Notes: This table reports estimated marginal effects from the ordered logit regression in column (4) of Table 4.

Table 6: Factors Influencing Approval with Conditions - Non-Delay Outcomes

Binary Logit Model
0=Approved with Conditions/Denied, 1=Approved

	(1)	(2)	(3)	(4)
Residential Development	0.202	0.225	0.470	0.188
	(0.380)	(0.391)	(0.413)	(0.436)
Mixed-Use Development	0.040	0.071	0.295	-0.053
	(0.383)	(0.398)	(0.420)	(0.452)
Non-Residential Development	-0.628	-0.672*	-0.543	-0.572
	(0.400)	(0.408)	(0.428)	(0.440)
ln(Square Footage)	-0.037	-0.033	-0.060	-0.089
	(0.073)	(0.076)	(0.080)	(0.081)
Height (ft)	-0.001	-0.001	-0.001	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)
$\log_2$(# Support)	-0.207***	-0.205***	-0.178**	-0.183**
	(0.074)	(0.076)	(0.081)	(0.082)
$\log_2$(# Oppose)	-0.073	-0.102	-0.159**	-0.167**
	(0.068)	(0.070)	(0.075)	(0.076)
Agenda Order	0.078	0.076	0.087	0.109*
	(0.056)	(0.059)	(0.062)	(0.063)
No. Agenda Items	0.041	0.049	0.091*	0.078
	(0.047)	(0.048)	(0.052)	(0.053)
Consent Calendar	1.147***	1.167***	1.253***	1.304***
	(0.286)	(0.296)	(0.304)	(0.306)
Atypicality	-0.146	-0.163	-0.137	-0.137
	(0.119)	(0.122)	(0.128)	(0.133)
Suffix Group Dummies	Y	Y	Y	Y
Council District Dummies	N	Y	Y	Y
Year Dummies	N	N	Y	Y
Embedding Cluster Dummies	N	N	N	Y
Observations	696	696	696	696
McFadden's Pseudo-R^2	0.083	0.102	0.144	0.149

Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Notes: This table reports coefficient estimates from the binary logit regression described in Section **??**, in which cases where delay was the outcome were removed from the estimation sample. The outcome is 1 if the case is approved, and 0 if the case is approved with conditions or denied. Dummies for missing values of height and square footage are included for cases without reported physical dimensions.

Table 7: Factors Influencing Approval or Delay - Re-hearings

Binary Logit
0=Delayed/Denied, 1=Approved/Approved with Conditions

	(1)	(2)	(3)	(4)	(5)
Atypicality	-0.172	0.060	0.416	-0.408	-0.050
	(0.392)	(0.477)	(0.547)	(0.455)	(0.407)
$1-$ Share Returning	-0.351	-0.525	-0.563	-0.642	-0.418
	(0.365)	(0.479)	(0.482)	(0.440)	(0.390)
Atypicality $\times$ ($1-$ Share Returning)	-0.740*	-1.125**	-1.010**	-0.734*	-0.729*
	(0.381)	(0.483)	(0.465)	(0.433)	(0.395)
ln(Days Since Last Hearing)	0.470	0.412	0.579	0.477	0.533*
	(0.293)	(0.357)	(0.370)	(0.313)	(0.307)
Appearances	-0.376	-0.244	0.060	-0.471*	-0.351
	(0.257)	(0.296)	(0.322)	(0.283)	(0.262)
Residential Development		-1.629			
		(2.533)			
Mixed-Use Development		-3.683			
		(2.698)			
Non-Residential Development		-3.866			
		(2.755)			
ln(Square Footage)		0.135			
		(0.273)			
Height (ft)		-0.001			
		(0.002)			
$\log_2$(# Support)				1.002***	
				(0.365)	
$\log_2$(# Oppose)				-0.299	
				(0.205)	
Agenda Order					0.308
					(0.250)
No. Agenda Items					-0.213
					(0.159)
Suffix Group Dummies	N	N	Y	N	N
Council District Dummies	Y	Y	Y	Y	Y
Year Dummies	Y	Y	Y	Y	Y
Embedding Cluster Dummies	Y	Y	Y	Y	Y
Observations	124	124	124	124	124
McFadden's Pseudo-R^2	0.282	0.348	0.390	0.345	0.301

Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Notes: This table reports coefficient estimates from the binary logit regression described in Section **??**, where the sample is restricted to re-hearings of previously heard cases. Dummies for missing values of height and square footage are included for cases without reported physical dimensions. Consent calendar is not included due to lack of variation. Atypicality and share returning are demeaned before estimation.

Table 8: What Happens to Delayed Cases?

Num Delayed Cases	80	
Num Resolved	73	(91.2%)
Avg. Time to Resolution	105 days	
Result		
Approved	24	(32.9%)
Approved with Minor Conditions	35	(47.9%)
Approved with Major Conditions	9	(12.3%)
Denied or Withdrawn	5	(6.8%)
Changes to Application		
No Changes	55	(75.3%)
Minor Changes	13	(17.8%)
Major Changes	5	(6.8%)

Notes: This table shows what happens to cases that get delayed in our data. Note that outcomes are tracked between the first observed hearing and the last observed hearing, though there may be additional hearings in between.

A Data Appendix

A.1 Downloading the Raw Data

The Los Angeles Planning Department keeps records of agenda, minutes, and supplemental documents for each City Planning Commission meeting. These are downloadable as PDF files. We wrote an automated scraper to crawl the Planning Department website (`https://planning.lacity.gov/about/commissions-boards-hearings#commissions`) and download these PDFs for each meeting. The first meeting date for which we obtained data was May 10th, 2018, and the last date was February 2, 2026. After the PDFs were downloaded, the textual content was extracted using the `pdfplumber` package in Python. Some pages were stored as images rather than searchable text; we used optical character recognition software (`tesseract`) to extract the text.

A.2 Extraction of Usable Fields

Agendas. We first extracted the unique agenda items out of the agenda text of each meeting. Because the agendas had a consistent format, we were able to do this using traditional pattern matching methods. For each agenda item, we used pattern matching to extract its agenda number (e.g. “1”, “2”, “3a”, “3b”, etc.) and the title (e.g. “Director’s Report and Commission Business”, “CPC-2014-2906-TDR-SPR”, etc.) We also used pattern matching to determine whether the agenda item is part of the consent calendar, as a consistent format was used to indicate such.

Once the agenda items were extracted from the agenda, we partitioned the agenda text into the section relevant for each item. We then asked `claude-opus-4-6` to extract the following information:[18]

- The address or location of the agenda item's proposed project, if applicable;
- The Council District of the proposed item;
- The Council Member in charge of said Council District;
- Whether or not the agenda item is an appeal of a previous decision;
- A short summary of the agenda item.

It should be noted that all this information is contained in the text of the agenda itself; the LLM does not need to rely on external knowledge. The model was instructed to return this data in a JSON.

Minutes. The minutes are much messier and inconsistently formatted than the agendas. This is because items can get moved around in the order in which they are discussed, or sometimes discussion begins on one item, is tabled temporarily, and discussion is resumed after other items were decided on.

To extract usable information from the minutes, we did the following: For each agenda item that was an active Planning Department case needing a decision by the CPC, we gave `claude-opus-4-6` the text of that agenda item and the *full text* of the minutes for that meeting. We then asked the model to provide:

- A one paragraph summary of the deliberations related to the agenda item and the final decision made by the Commission;
- If a motion was made, the name of the Commission Member who made the motion;

[18] The full prompts used are available at `https://github.com/ed-kung/lacpc`.

- The name of the seconder;
- The list of Members who voted yes, voted no, abstained, were absent, or were recused;
- The result of the vote (passed or failed);
- The implication of the vote for the proposed project (approved, approved with minor conditions, approved with major conditions, delayed, or denied).

Sometimes multiple motions are made on the same agenda item. This usually happens if the first motion fails. In such cases, the model was instructed to take only the result of the final motion. An advantage of using a state-of-the-art model like `claude-opus-4-6` is that it was smart enough to properly contextualize the implication of the vote for the proposed project. For example, the model understood that a unanimous vote to reject an appeal implies that the project itself is approved, or that a vote to continue the deliberations to a future date implies that the decision on the project is delayed.

Supplemental Documents. The supplemental documents were challenging to process because for each meeting, all supplemental documents submitted to that meeting were spliced into a single PDF. This resulted in PDFs of many hundreds of pages, containing potentially hundreds of separate documents. In our experimentation, even state-of-the-art LLMs had trouble identifying the document boundaries, especially when there were many pages of documents, or when a document contained many pages of attachments, or a page of signatures. We therefore went through the supplemental PDFs manually to identify the document boundaries. After manually identifying and extracting the individual documents out of the spliced PDF, we did the following: For each supplemental document submitted to each meeting, we gave `claude-opus-4-6` the text of the supplemental document and the *full agenda text* of that meeting, then asked it to provide:

- A short summary of the supplemental document and its relation to any items on the agenda;
- A list of the agenda items that the supplemental document references, identifying the items by their item number;
- Document type (public comment, supplemental materials, administrative or other);
- Support or oppose: For each of the referenced agenda items, the model was asked the say whether the document definitely support, somewhat supports, is neutral towards, somewhat opposes, or definitely opposes the project being proposed.

As with the minutes, proper contextualization of the document with respect to the referenced agenda items is crucial. For example, a public comment may support the appeal of a decision, which would be an opposition to the underlying project. A pattern-matching approach or a less intelligent model may see words like "support" and think that the letter was written in support of the project, despite the truth being quite the opposite. This again highlights the importance of using highly capable models like `claude-opus-4-6` to handle the extraction of the data.

Table A1: Case Suffix Meanings and Groupings

Suffix	Meaning	Count	Description	Group
SPR	Site Plan Review	264	Indicates that project requires Site Plan Review, a discretionary review required for qualifying projects	Site Plan Review
1A	Appeal (1st)	259	Indicates that an appeal has been filed against a prior determination in the case	Appeals
DB	Density Bonus	247	Indicates that the project is using the city's Density Bonus Program which offers density incentives in exchange for providing affordable housing	Incentive Programs
CU	Conditional Use	192	Indicates that the applicant seeks a Conditional Use Permit	Conditional Use Permits
HCA	Housing Crisis Act	153	Indicates that the project is subject to the California Housing Crisis Act, which limits local discretion over qualifying housing projects	Incentive Programs
TOC	Transit Oriented Communities	122	Indicates that the project is using the city's Transit Oriented Communities Program which offers incentives to qualifying projects located near public transit	Incentive Programs
HD	Height District	88	Indicates that the applicant seeks to change the project's applicable Height District	Variances/Adjustments/Exceptions
SPP	Specific Plan Project Permit	85	Indicates that the project requires a permit for development under an applicable Specific Plan	Compliance Review
MCUP	Master Conditional Use Permit	74	Indicates that the applicant requests a Master Conditional Use Permit	Conditional Use Permits
ZC	Zone Change	70	Indicates that the applicant seeks to change the project's zoning designation	Variances/Adjustments/Exceptions
GPA	General Plan Amendment	67	Indicates that the applicant seeks an amendment to the General Plan	Code or Plan Amendments
VZC	Vesting Zone Change	65	Indicates that the applicant seeks a zone change with vesting protections	Variances/Adjustments/Exceptions
WDI	Waiver of Dedication and Improvements	58	Indicates that the applicant is seeking relief from required right-of-way dedication or street improvement requirements	Development Agreements
VHCA	Vesting Housing Crisis Act	57	Indicates that the project is utilizing the Housing Crisis Act and is seeking vesting protections	Incentive Programs
CA	Code Amendment	44	Indicates that the applicant seeks an amendment to applicable zoning code	Code or Plan Amendments
CUB	Conditional Use Beverage	29	Indicates that the applicant seeks a Conditional Use Permit for serving alcoholic beverages	Conditional Use Permits
PHP	Priority Housing Project	27	Indicates that the project is eligible for priority processing under the city's Priority Housing Program	Incentive Programs
ZV	Zone Variance	27	Indicates that the applicant seeks relief from specified zoning standards based on special circumstances or hardship	Variances/Adjustments/Exceptions
TDR	Transfer of Development Rights	26	Indicates that the case involves transferring unused development capacity from one site to another	Development Agreements
SP	Specific Plan	25	Indicates that the case seeks adoption of a new Specific Plan or amendment of an existing one	Code or Plan Amendments
ZAA	Zoning Administrator Adjustment	23	Indicates that the applicant seeks limited adjustments to dimensional standards such as yard, setback, area, height, building line, and floor area rules	Variances/Adjustments/Exceptions
DRB	Design Review Board	22	Indicates that the project requires Design Review Board review	Compliance Review
GPAJ	General Plan Amendment Measure JJJ	20	Indicates that the applicant seeks a General Plan amendment subject to Measure JJJ affordable housing and labor standards	Code or Plan Amendments
CDP	Coastal Development Permit	18	Indicates that the applicant seeks a Coastal Development Permit	Compliance Review
SN	Sign District	17	Indicates that the case seeks adoption of a new Sign District or amendment of an existing one	Code or Plan Amendments
VCU	Vesting Conditional Use	17	Indicates that the applicant seeks a Conditional Use Permit with vesting protections	Conditional Use Permits
VZCJ	Vesting Zone Change Measure JJJ	17	Indicates that the applicant seeks a zone change with vesting protections subject to Measure JJJ	Variances/Adjustments/Exceptions
SIP	Streamlined Infill Process	15	Indicates the applicant seeks streamlined review under California SB 35, which streamlines review for qualified infill development projects	Incentive Programs
CU3	Class 3 Conditional Use	14	Indicates that the applicant requests a Class 3 Conditional Use Permit, which is more substantial in scope than other Conditional Use Permits	Conditional Use Permits
BL	Building Line	14	Indicates that the applicant seeks to establish, modify, or remove a building line	Variances/Adjustments/Exceptions
ZAD	Zoning Administrator Determination	14	Indicates that the applicant seeks a special determination or waiver from a Zoning Administrator	Other
CUX	Conditional Use Adult Entertainment	13	Indicates that the applicant seeks a Conditional Use Permit for adult entertainment	Conditional Use Permits
CDO	Community Design Overlay	13	Indicates that the project requires review for compliance with an applicable Community Design Overlay District	Compliance Review
CN	Condominium	13	Indicates that the project is a condominium project	Other
MEL	Mello Act Compliance Review	12	Indicates that the project requires Mello Act compliance review	Compliance Review
PA1	Plan Approval (1st)	11	Indicates the project requires a Plan Approval determination, typically to confirm consistency with a prior planning decision	Conditional Use Permits
ZCJ	Zone Change Measure JJJ	11	Indicates that the applicant seeks a zone change subject to Measure JJJ	Variances/Adjustments/Exceptions
DD	Director's Determination	11	Indicates that the project requires discretionary determination by the Director of Planning	Other

Notes: This table shows all Planning Department case suffixes present in the dataset, along with their meanings, counts, and assigned groups. The suffixes are grouped into categories based on their meaning and usage.

Table A2: Case Suffix Meanings and Groupings (cont'd)

Suffix	Meaning	Count	Description	Group
PR	Project Review	10	Functionally equivalent to Site Plan Review	Site Plan Review
DA	Development Agreement	9	Indicates that the applicant seeks approval of a development agreement with the City	Development Agreements
SPE	Specific Plan Exception	9	Indicates that the applicant seeks an exception to the applicable Specific Plan	Variances/Adjustments/Exceptions
BSA	Building and Safety Appeal	9	Indicates that an appeal has been filed against a prior determination by the Department of Building and Safety	Appeals
SPPC	Specific Plan Project Compliance	8	Indicates that the project requires review for compliance with an applicable Specific Plan	Compliance Review
RDP	Redevelopment Plan Project	8	Indicates that the project requires review for compliance with an applicable Redevelopment Plan	Compliance Review
MSC	Miscellaneous	8	Indicates that the case involves miscellaneous planning actions	Other
F	Fence Height	7	Indicates that the applicant seeks an entitlement related to fence height	Variances/Adjustments/Exceptions
M3	Modification (3rd)	7	Indicates that a third modification request has been filed for a prior planning approval	Development Agreements
MSP	Mulholland Specific Plan	6	Indicates that the project requires review under the Mulholland Specific Plan	Compliance Review
PSH	Permanent Supportive Housing	6	Indicates that the project involves Permanent Supportive Housing or seeks related incentives or review	Incentive Programs
CPIOA	Community Plan Implementation Overlay Adjustment	5	Indicates that the applicant seeks an adjustment from standards in an applicable Community Plan Implementation Overlay District	Variances/Adjustments/Exceptions
CPU	Community Plan Update	5	Indicates that the case is part of a Community Plan Update	Code or Plan Amendments
PMLA	Preliminary Parcel Map	5	Indicates that the applicant seeks approval of a preliminary parcel map for a subdivision or condominium project	Other
CCMP	Certificate of Compatibility	5	Indicates that the project requires a Certificate of Compatibility for work in an applicable district or regulated area	Other
SPPA	Specific Plan Project Permit Adjustment	5	Indicates that the applicant seeks an adjustment from standards in an applicable Specific Plan Project Permit process	Variances/Adjustments/Exceptions
M1	Modification (1st)	4	Indicates that a first modification request has been filed for a prior planning approval	Development Agreements
SUD	Supplemental Use District	4	Indicates that the case involves adoption, amendment, or review under a Supplemental Use District	Code or Plan Amendments
ZAI	Zoning Administrator Interpretation	4	Indicates that the applicant seeks an interpretation from the Zoning Administrator	Other
PUB	Public Benefit	4	Indicates that the project involves a Public Benefit approval or review	Incentive Programs
CUW	Conditional Use Wireless	3	Indicates that the applicant seeks a Conditional Use Permit for a wireless telecommunications facility	Conditional Use Permits
SL	Small Lot Subdivision	3	Indicates that the project involves a small lot subdivision	Other
DI	Director of Planning Interpretation	2	Indicates that the applicant seeks an interpretation from the Director of Planning	Other
PMEX	Parcel Map Exemption	2	Indicates that the applicant seeks an exemption from parcel map requirements	Other
SPPE	Specific Plan Project Exception	2	Indicates that the applicant seeks an exception from standards in an applicable Specific Plan Project review process	Variances/Adjustments/Exceptions
ELD	Elder Care Facilities	2	Indicates that the project involves an elder care facility approval or review	Conditional Use Permits
CPIO	Community Plan Implementation Overlay	2	Indicates that the case seeks adoption or amendment of a Community Plan Implementation Overlay District	Code or Plan Amendments
PAB	Plan Approval Booze	2	Indicates that the applicant seeks a Plan Approval related to a prior alcohol-related approval	Conditional Use Permits
SCEA	Sustainable Communities Environmental Assessment	1	Indicates that environmental review is being processed as a Sustainable Communities Environmental Assessment	Other
GB	Green Building	1	Indicates that the case involves Green Building review or requirements	Compliance Review
ACI	Amendment to Council Instructions	1	Indicates that the case involves an amendment to prior Council instructions	Development Agreements
COA	Certificate of Appropriateness	1	Indicates that the project requires a Certificate of Appropriateness, typically for work affecting a historic resource or district	Compliance Review
AMDT2	Amendment (2nd)	1	Indicates that a second amendment request has been filed for a prior planning approval	Development Agreements
AMDT1	Amendment (1st)	1	Indicates that a first amendment request has been filed for a prior planning approval	Development Agreements
PA2	Plan Approval (2nd)	1	Indicates the project requires a second Plan Approval determination, typically to confirm consistency with a prior planning decision	Conditional Use Permits
AMDT3	Amendment (3rd)	1	Indicates that a third amendment request has been filed for a prior planning approval	Development Agreements
AD	Annexation/Detachment	1	Indicates that the case involves annexation or detachment of land	Other
CU2	Class 2 Conditional Use	1	Indicates that the applicant requests a Class 2 Conditional Use Permit	Conditional Use Permits
RDPA	Redevelopment Plan Project Adjustment	1	Indicates that the applicant seeks an adjustment related to an applicable Redevelopment Plan Project review	Variances/Adjustments/Exceptions

Notes: This table shows all Planning Department case suffixes present in the dataset, along with their meanings, counts, and assigned groups. The suffixes are grouped into categories based on their meaning and usage.